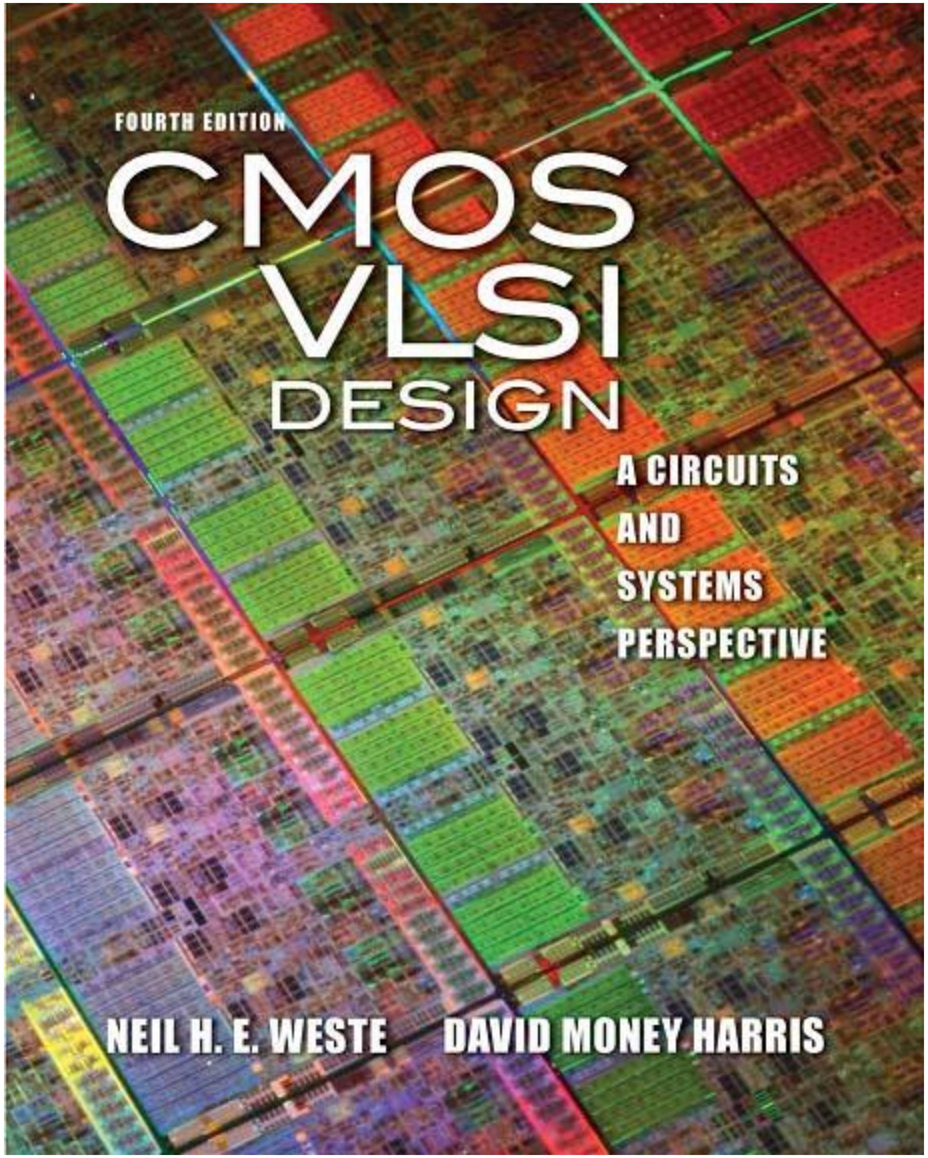
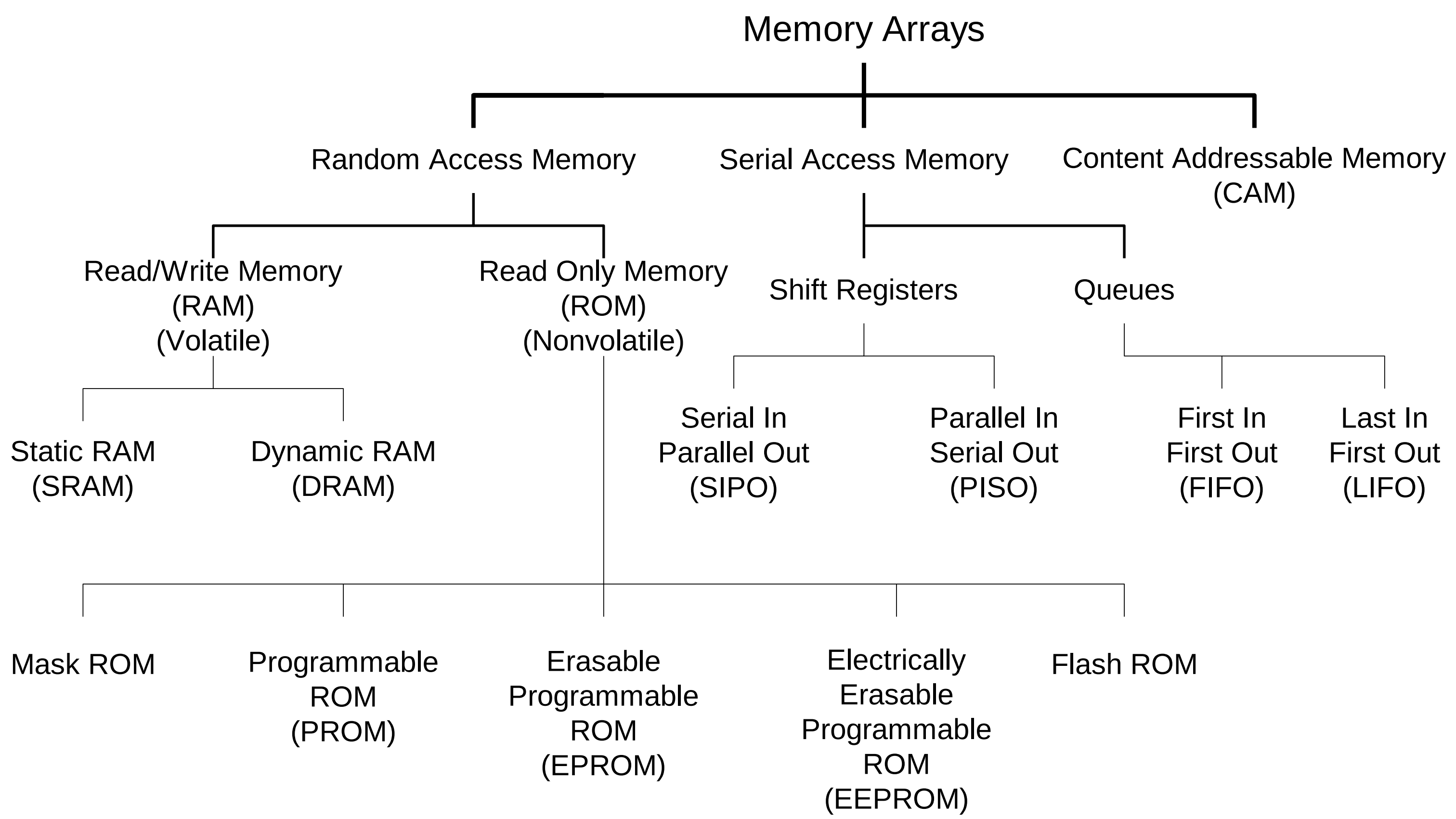


Array Subsystems

Lecture # 1

- Memory Arrays
- SRAM Architecture
  - SRAM Cell
  - SRAM Read
  - SRAM Write
  - SRAM Sizing
  - SRAM Column for Read & Write
- Decoders
- Column Circuitry
  - Bitline conditioning
  - Sense amplifiers
  - Column multiplexing
- Multiple Ported SRAM

Memory Arrays

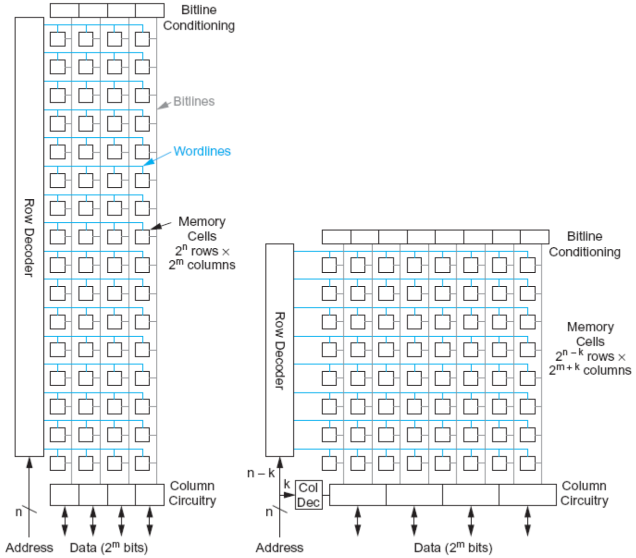


Volatile Memories

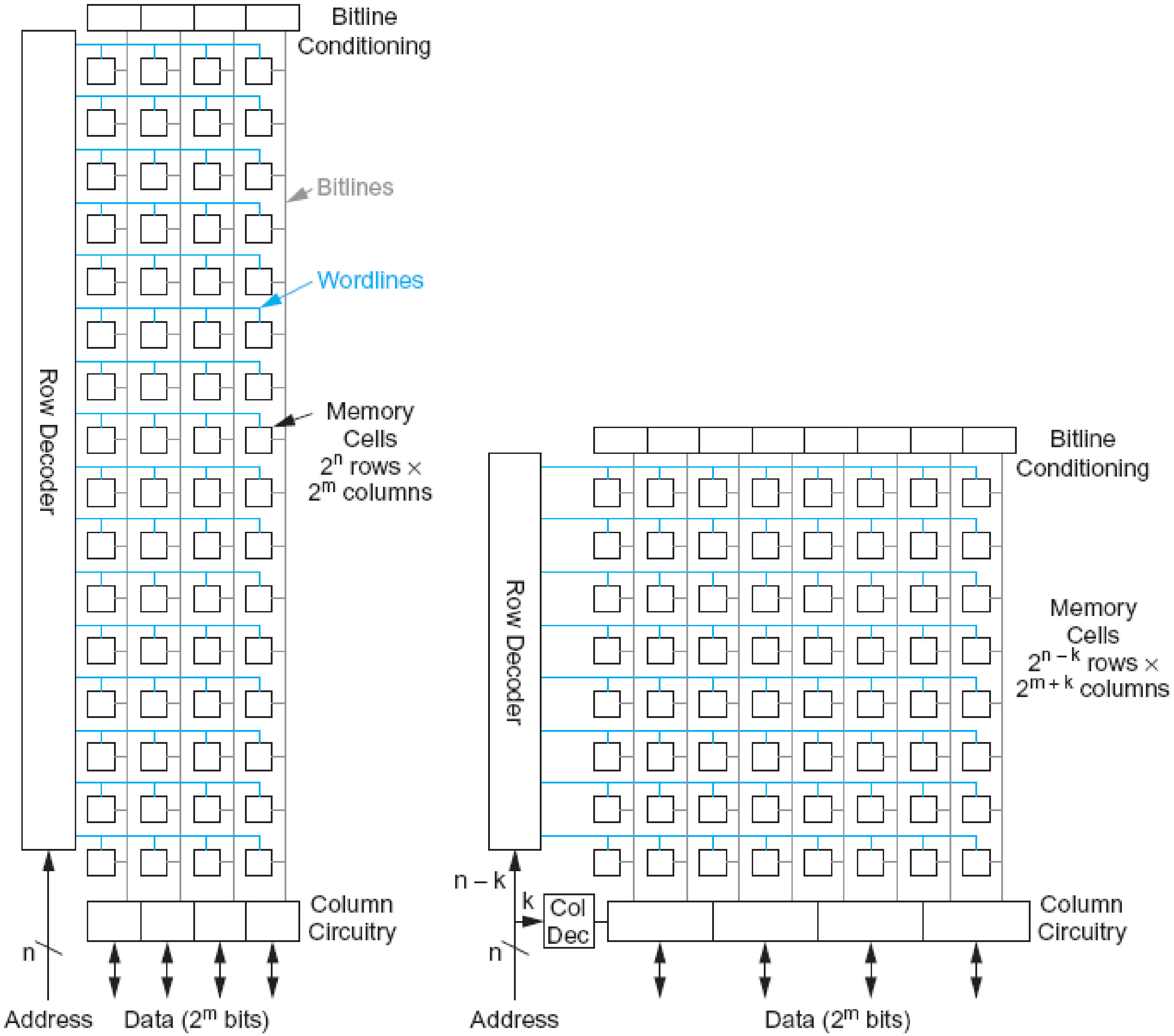
- Memory cells used in volatile memories are static structures and dynamic structures: SRAM & DRAM.
- Static cells use feedback to maintain their state, while dynamic cells use charge stored on a floating capacitor through an access transistor.
- Charge will leak away through the access transistor even while the transistor is OFF, so dynamic cells must be periodically read and rewritten to refresh their state.
- SRAMs are faster and less troublesome, but require more area per bit than DRAMs.

Array Architecture

- 2^n *words* of 2^m *bits* each
- If n >> m, fold by 2^k into fewer *rows* of more *columns*



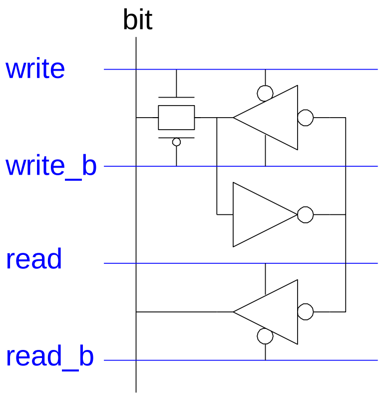
- Good regularity – easy to design
- Very high density if good cells are used

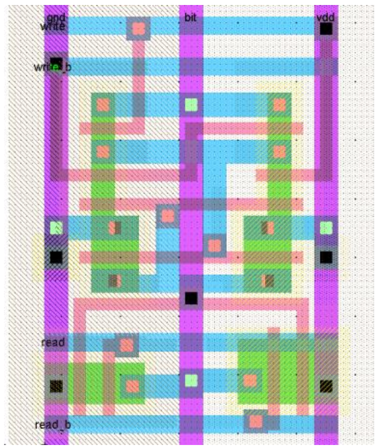


Bitline Conditioning
Bitlines
Wordlines
Memory Cells
2^n rows × 2^m columns
Row Decoder
Column Circuitry
n
Address
Data (2^m bits)
Bitline Conditioning
Memory Cells
2^(n − k) rows × 2^(m + k) columns
Row Decoder
n – k
k
Col Dec
n
Column Circuitry
Address
Data (2^m bits)

12T SRAM Cell

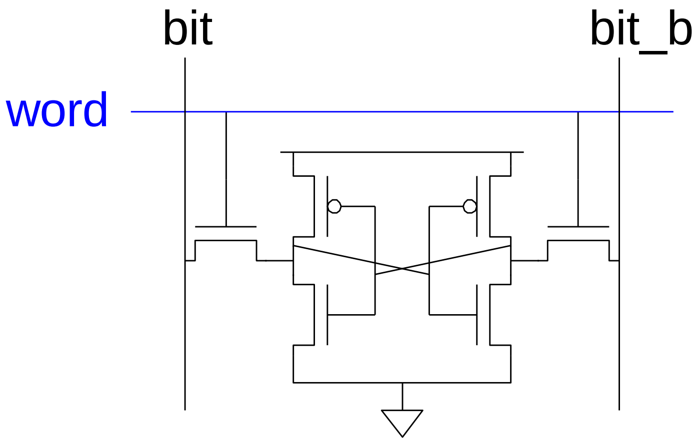
- Basic building block: SRAM Cell
  - Holds one bit of information, like a latch
  - Must be read and written
- 12-transistor (12T) SRAM cell
  - Use a simple latch connected to bitline
  - 46 x 75 λ unit cell





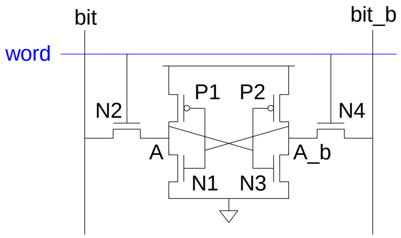
6T SRAM Cell

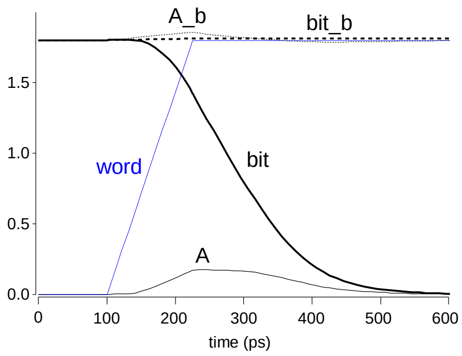
- Cell size accounts for most of array size
  - Reduce cell size at expense of complexity
- 6T SRAM Cell
  - Used in most commercial chips
  - Data stored in cross-coupled inverters
- Read:
  - Precharge bit, bit_b
  - Raise wordline
- Write:
  - Drive data onto bit, bit_b
  - Raise wordline



SRAM Read

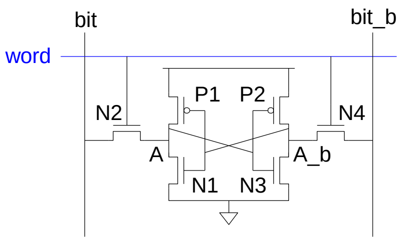
- Precharge both bitlines high
- Then turn on wordline
- One of the two bitlines will be pulled down by the cell
- Ex: A = 0, A_b = 1
  - bit discharges, bit_b stays high
  - But A bumps up slightly
- *Read stability*
  - A must not flip
  - N1 >> N2

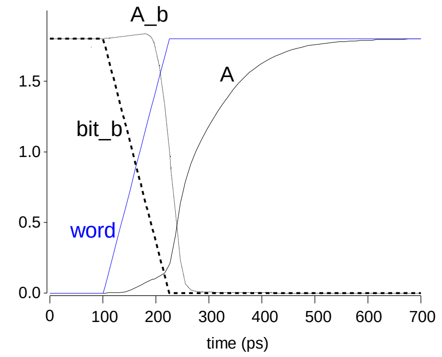




SRAM Write

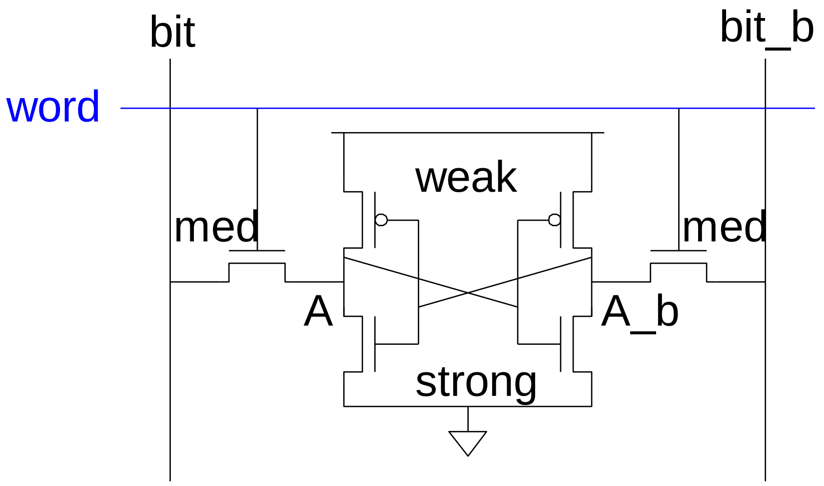
- Drive one bitline high, the other low
- Then turn on wordline
- Bitlines overpower cell with new value
- Ex: A = 0, A_b = 1, bit = 1, bit_b = 0
  - Force A_b low, then A rises high
- *Writability*
  - Must overpower feedback inverter
  - P2 << N4





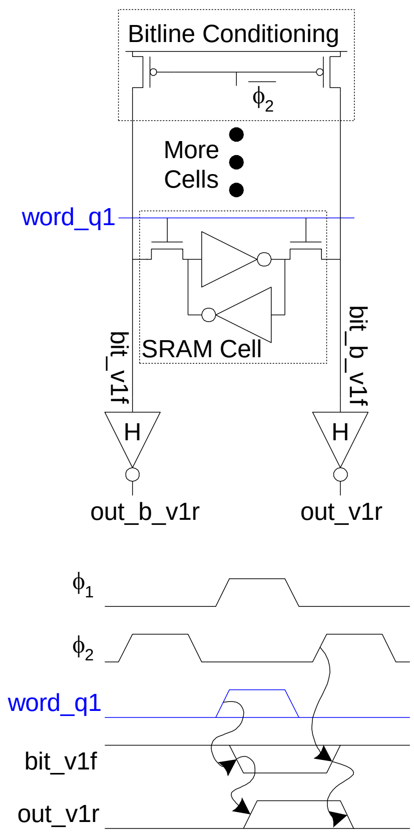
SRAM Sizing

- High bitlines must not overpower inverters during reads
- But low bitlines must write new value into cell

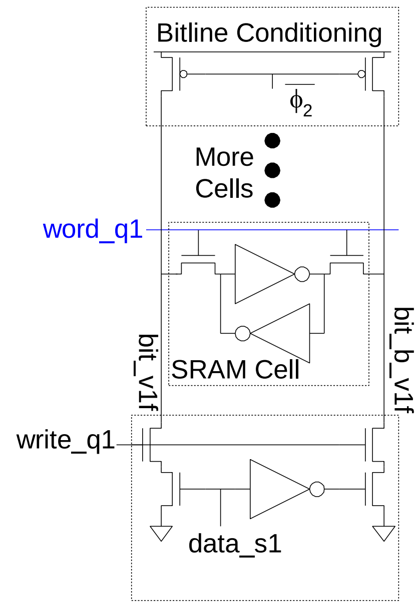


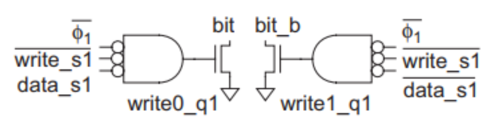
SRAM Column Example

Read



Write





(b)

FIGURE 12.7 SRAM column write

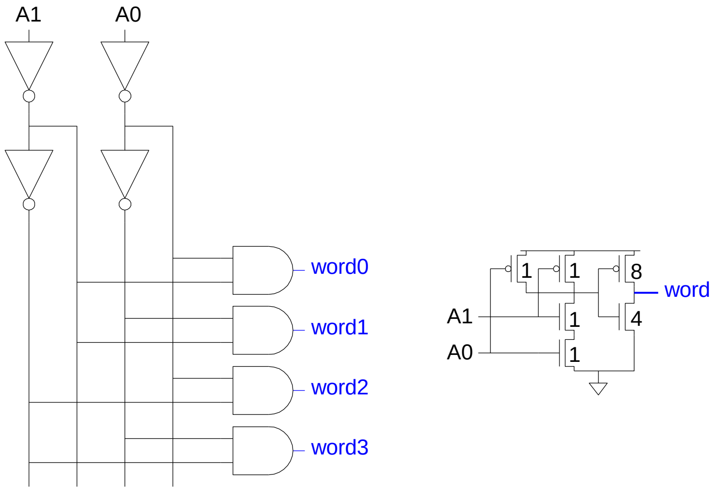
Home Work

- Design a 2 kwords by 16 columns SRAM 8-way folded into 256 rows by 128 columns at block level with its supporting circuitry.
- Specifically identify how many decoders and multiplexers will be used and what are the sizes of these circuits.

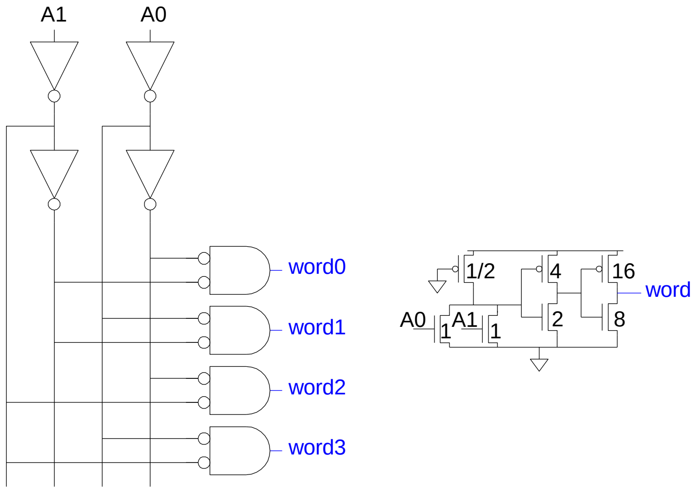
Decoders

- n:2^n decoder consists of 2^n n-input AND gates
  - One needed for each row of memory
  - Build AND from NAND or NOR gates

Static CMOS

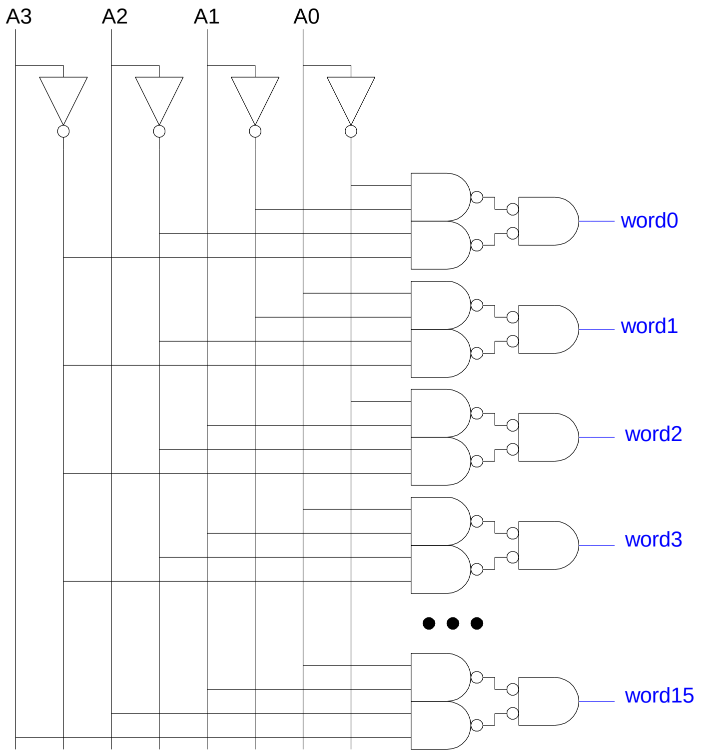


Pseudo-nMOS



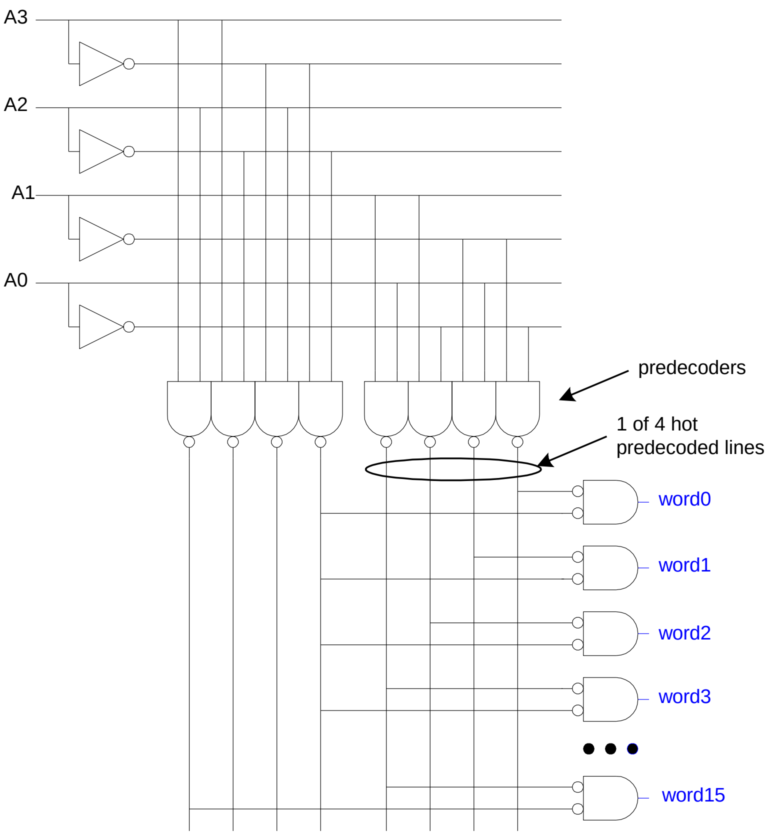
Large Decoders

- For n > 4, NAND gates become slow
  - Break large gates into multiple smaller gates



Predecoding

- Many of these gates are redundant
  - Factor out common gates into predecoder
  - Saves area
  - Same path effort

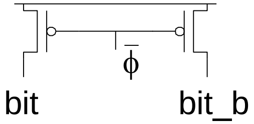


Column Circuitry

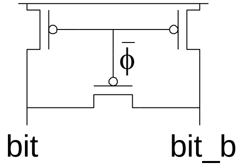
- Some circuitry is required for each column
  - Bitline conditioning
  - Sense amplifiers
  - Column multiplexing

Bitline Conditioning

- Precharge bitlines high before reads



- Equalize bitlines to minimize voltage difference when using sense amplifiers

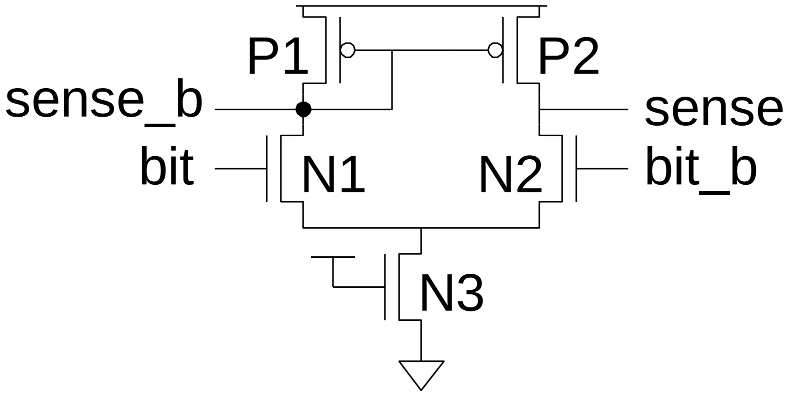


Sense Amplifiers

- Bitlines have many cells attached
  - Ex: 32-kbit SRAM has 128 rows x 256 cols
  - 128 cells on each bitline
- $t_{pd} \propto (C/I)\ \Delta V$
  - Even with shared diffusion contacts, 64C of diffusion capacitance (big C)
  - Discharged slowly through small transistors (small I)
- *Sense amplifiers* are triggered on small voltage swing (reduce ΔV)

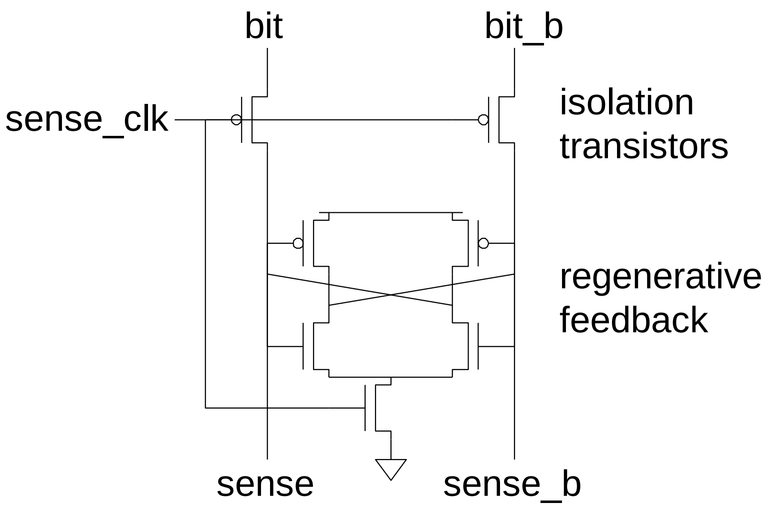
Differential Pair Amp

- Differential pair requires no clock
- But always dissipates static power



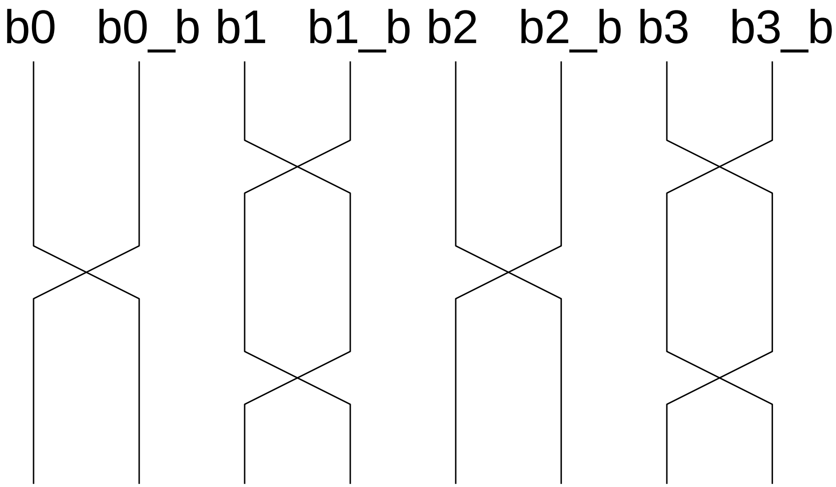
Clocked Sense Amp

- Clocked sense amp saves power
- Requires sense_clk after enough bitline swing
- Isolation transistors cut off large bitline capacitance



Twisted Bitlines

- Sense amplifiers also amplify noise
  - Coupling noise is severe in modern processes
  - Try to couple equally onto bit and bit_b
  - Done by *twisting* bitlines

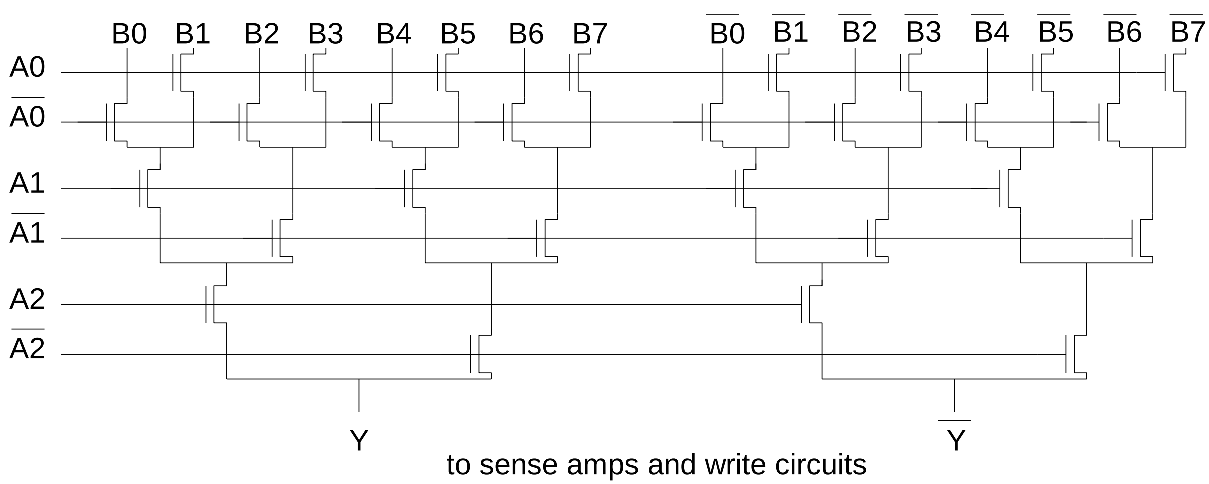


Column Multiplexing

- Recall that array may be folded for good aspect ratio
- Ex: 2 kword x 16 folded into 256 rows x 128 columns
  - Must select 16 output bits from the 128 columns
  - Requires 16 8:1 column multiplexers

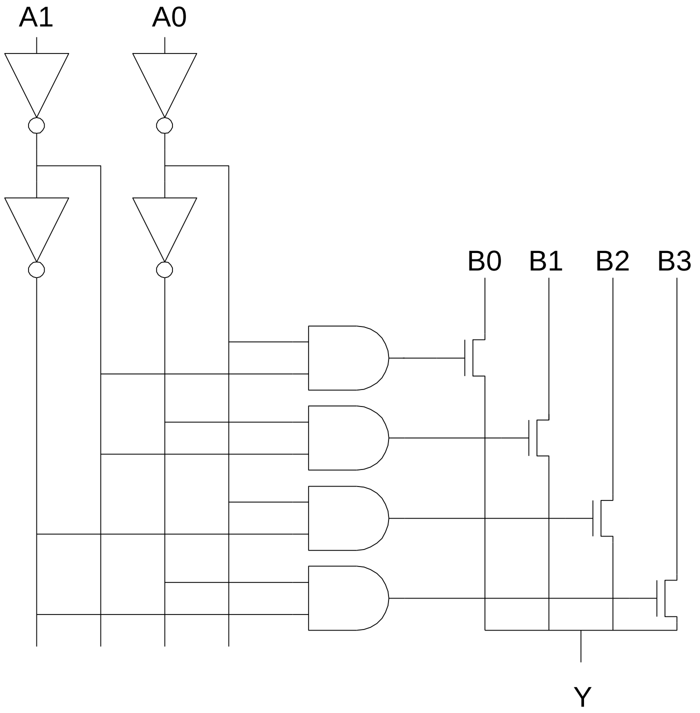
Tree Decoder Mux

- Column mux can use pass transistors
  - Use nMOS only, precharge outputs
- One design is to use k series transistors for 2^k:1 mux
  - No external decoder logic needed

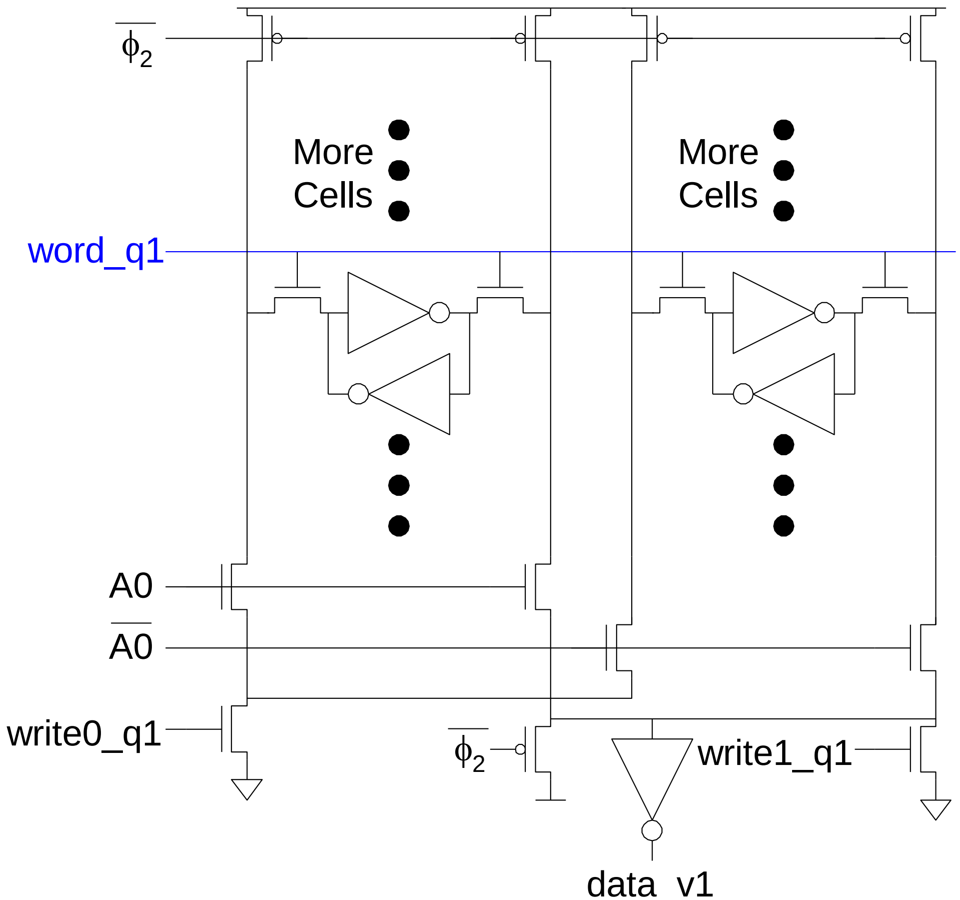


Single Pass-Gate Mux

- Or eliminate series transistors with separate decoder



Ex: 2-way Muxed SRAM

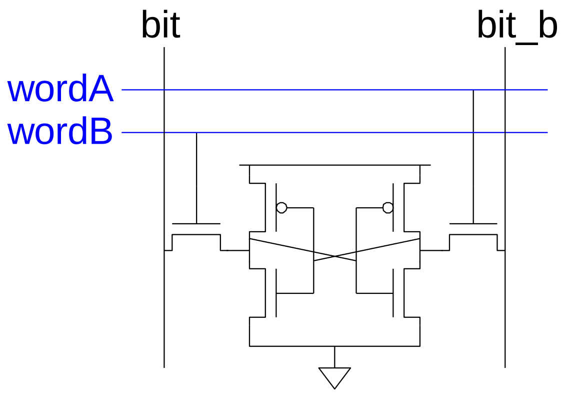


Multiple Ports

- We have considered single-ported SRAM
  - One read or one write on each cycle
- *Multiported* SRAM are needed for register files
- Examples:
  - Multicycle MIPS must read two sources or write a result on some cycles
  - Pipelined MIPS must read two sources and write a third result each cycle
  - Superscalar MIPS must read and write many sources and results each cycle

Dual-Ported SRAM(6T)

- Simple dual-ported SRAM
  - Two independent single-ended reads
  - Or one differential write



- Do two reads and one write by time multiplexing
  - Read during ph1, write during ph2

More Dual-Ported SRAM

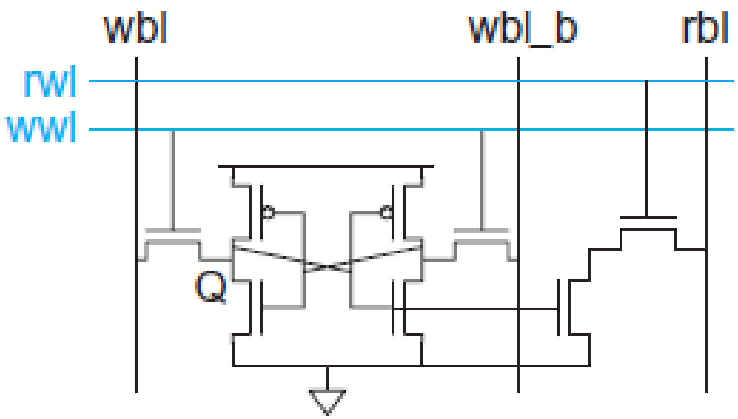


FIGURE 12.18 8T dual-port SRAM cell

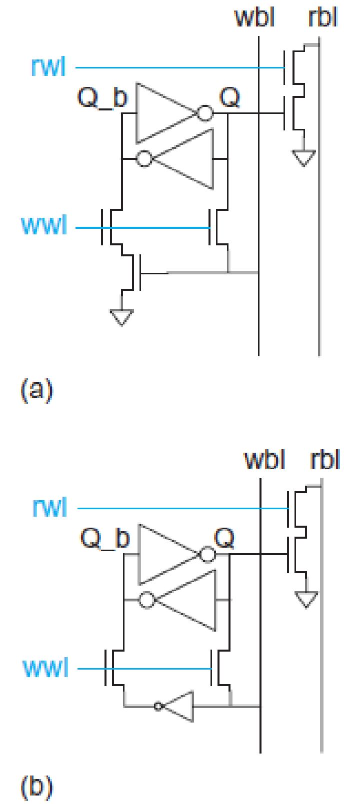
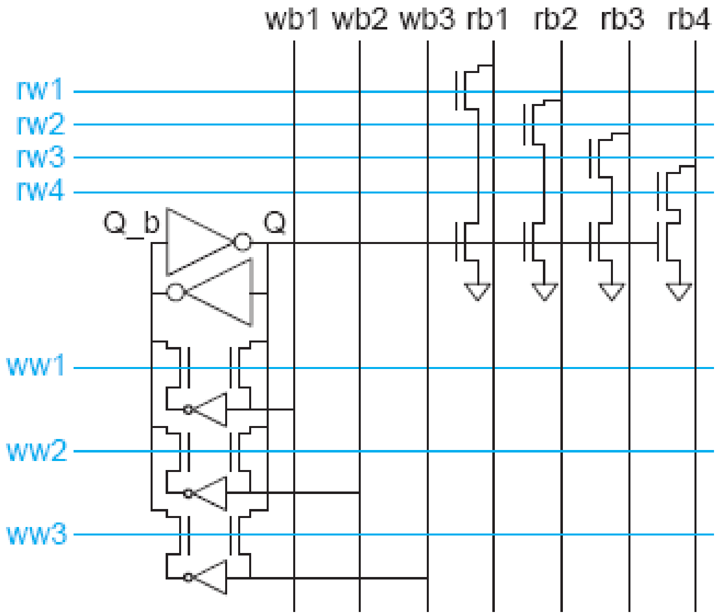


FIGURE 12.33 Register cell with single-ended write port

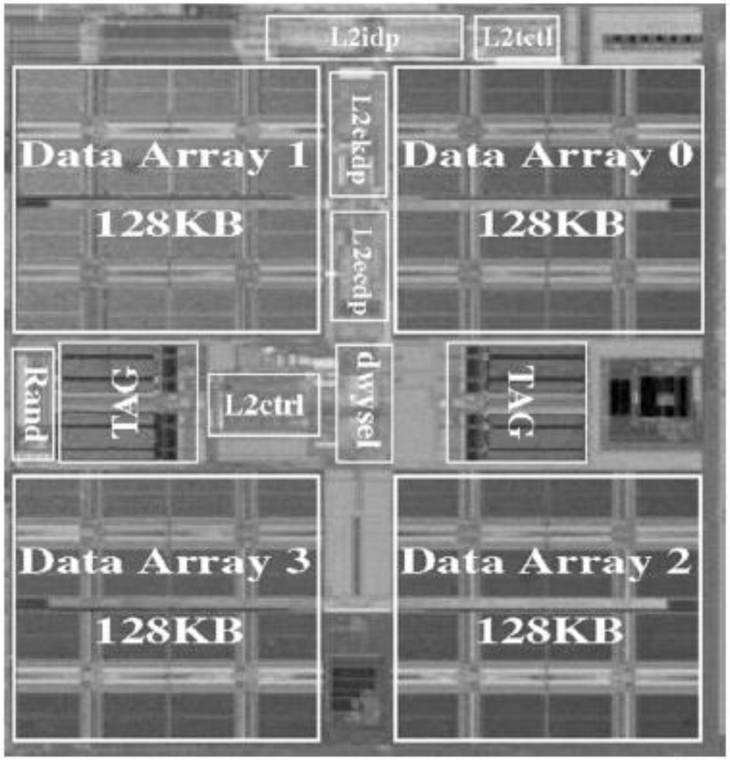
Multi-Ported SRAM

- Adding more access transistors hurts read stability
- Multiported SRAM isolates reads from state node
- Single-ended bitlines save area



Large SRAMs

- Large SRAMs are split into subarrays for speed
- Ex: UltraSparc 512KB cache
  - 4 128 KB subarrays
  - Each have 16 8KB banks
  - 256 rows x 256 cols / bank
  - 60% subarray area efficiency
  - Also space for tags & control



[Shin05]

Home Work

- Design a 4 rows by 2 columns SRAM at transistor level with their supporting circuitry.